Final Project Report MATH 563

Alexandre St-Aubin
Student ID: 261115785

Jonathan Campana
Student ID: 260985613

Edmund Wu
Student ID:

Elliot Forcier-Poirier
Student ID:

Abstract

give some background

1 Introduction

1.1 Notation

Throughout this paper, we will denote the *proximal operator* of f with parameter λ by

$$\mathbf{prox}_{\lambda f}(v) := \arg\min_{x} \left\{ f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right\} \tag{1}$$

The *indicator function* for a convex set C will be defined as

$$\delta_C(x) := \begin{cases} 0, & \text{if } x \in C \\ +\infty, & \text{if } x \notin C \end{cases} \tag{2}$$

We also denote the *Euclidean Projection* on a set C by

$$\mathbf{P}_C(v) := \arg\min_{x \in C} \|x - v\|_2 \tag{3}$$

1.2 Preleminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \tag{4}$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f. In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator)**.** A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in R^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \tag{5}$$

Preprint. Under review.

Proposition 3 (Proximal operator splitting)**.** If f is separable across two variables, so $f(x,y) = \varphi(x) + \psi(y)$, then

$$\mathbf{prox}_f(x,y) = (\mathbf{prox}_\varphi(x), \mathbf{prox}_\psi(y)) \tag{6}$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

Proposition 4. (Proximal operator of norms) If $f = \|\cdot\|$ is a general norm, then f^* is the indicator function on the unit ball for the dual norm $\|\cdot\|_*$, namely

$$\|\cdot\|^* = \delta_{\|\cdot\|_* \leq 1}$$

where $\|\cdot\|_*$ is given by

$$\|v\|_* = \sup_x \{\langle v, x\rangle \mid \|x\| \leq 1\}$$

In particular, the *Moreau Decomposition Theorem* gives that

$$\mathbf{prox}_{\|\cdot\|}(v) = \mathbf{P}_{\{x\,:\,\|x\|_* \leq 1\}}(v) - v$$

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{aligned} &\min_{x,y} f(x) + g(y) \\ &\text{subject to } Ax = y, \end{aligned} \tag{7}$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b, along with the kernel K used to blur it using the model

$$Kx + \eta = b, \tag{8}$$

where x is the original image, and η the added noise. Our goal is to recover the original image x. We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$\min_x \|Kx - b\|_1 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{9}$$

and second,

$$\min_x \|Kx - b\|_2^2 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{10}$$

where $S = \{x \mid 0 \leq x \leq 1\}$, so that δ_S aims to restrict the range of allowed pixels. The iso norm is defined as

$$\|(x,y)\|_{\text{iso}} = \sum_{k=1}^{n} (x_k^2 + y_k^2)^{\frac{1}{2}}$$

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (9), let

$$\begin{aligned} f(x) =& \delta_S(x) \\ g(x) =& \|Kx - b\|_1 + \gamma\|Dx\|_{\text{iso}} \end{aligned} \tag{11}$$

and consider the minimization problem

$$\min_{x,y} f(x) + g(y) + \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) \tag{12}$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}\right) =: \mathcal{F}(x,y) \tag{13}$$

As stated in the literature [2], $\mathcal{F}$ is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (5), the zeros of $\mathcal{F}$ are the fixed points of the resolvent of $\mathcal{F}$. Therefore, if we can find the fixed points of $\mathcal{F}$, we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split $\mathcal{F}$ into two maximal monotone operators $\mathcal{A}$ and $\mathcal{B}$ which have inexpensive proximal operators. Finding the fixed points of $\mathcal{A}+\mathcal{B}$ will be equivalent to finding that of $\mathcal{F}$. Define $\mathcal{A}$ and $\mathcal{B}$ as,

$$\begin{aligned} \mathcal{A}(x,y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky-b\|_1 + \gamma\|Dy\|_{\text{iso}}) \\ \mathcal{B}(x,y) &= \partial\delta_0\left(\begin{bmatrix}\begin{bmatrix}K\\D\end{bmatrix} & -I\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}\right) \end{aligned} \tag{14}$$

We now focus on finding the resolvents of $\mathcal{A}$ and $\mathcal{B}$,

$$\begin{aligned} J_{\mathcal{A}} = (I+\lambda\mathcal{A})^{-1} &= \mathbf{prox}_{\lambda(f+g)}(x,y) \quad \text{[by (4)]} \\ &= (\mathbf{prox}_{\lambda f}(x), \mathbf{prox}_{\lambda g}(y)) \quad \text{[by Prop. 3]} \end{aligned} \tag{15}$$

where

$$\mathbf{prox}_{\lambda f}(x) = \mathbf{prox}_{\lambda\delta_S}(x) = \arg\min_u \frac{1}{2}\|x-u\|_2^2 + \delta_S(x) = \arg\min_{u\in S}\frac{1}{2}\|x-u\|_2^2 = \mathbf{P}_S(x), \tag{16}$$

which is the euclidean projection operator onto S. It can be computed element-wise by

$$(\mathbf{P}_S(x))_i = \begin{cases} 0, & \text{if } x_i < 0 \\ x_i, & \text{if } x_i \in S \\ 1, & \text{if } x_i > 1 \end{cases} = \max\{0, \min\{x_i, 1\}\} \tag{17}$$

Now, we observe that $g(y)$ described in (11) can be written in a different form by letting

$$g(y_1,y_2,y_3) = \|y_1-b\|_1 + \gamma\|(y_2,y_3)\|_{\text{iso}} \tag{18}$$

subject to

$$y_1 = Ky \quad \text{and} \quad \begin{bmatrix}y_2\\y_3\end{bmatrix} = Dy \iff \begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}K\\D\end{bmatrix}y$$

Then, by Proposition 3, notice that the proximal operator of g may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1-b\|_1 \quad \text{and} \quad \psi(y_2,y_3) = \gamma\|(y_2,y_3)\|_{\text{iso}}$$

which gives

$$\mathbf{prox}_{\lambda g}(y_1,y_2,y_3) = (\mathbf{prox}_{\lambda\varphi}(y_1), \mathbf{prox}_{\lambda\psi}(y_2,y_3)) \tag{19}$$

We know that the proximal operator of the L^1 norm is the soft-thresholding operator T_λ, so $\mathbf{prox}_{\lambda\varphi}(y_1)$ is given element-wise as follows,

$$(\mathbf{prox}_{\lambda\varphi}(y_1))_i = (\mathbf{prox}_{\lambda\|\cdot\|_1}(y_1-b))_i = T_\lambda((y_1-b)_i) := \begin{cases} (y_1-b)_i - \lambda, & \text{if } (y_1-b)_i > \lambda \\ (y_1-b)_i + \lambda, & \text{if } (y_1-b)_i < -\lambda \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

As a result of Proposition 4, $\mathbf{prox}_{\lambda\psi}(y_2,y_3)$, is given element-wise by

$$(\mathbf{prox}_{\lambda\psi}(y_2,y_3))_i = \alpha_i((y_2)_i,(y_3)_i),$$

where,

$$\alpha_i = \begin{cases} 1 - \frac{\lambda}{\sqrt{(y_2)_i^2+(y_3)_i^2}}, & \text{if } \sqrt{(y_2)_i^2+(y_3)_i^2} > \lambda \\ 0, & \text{otherwise.} \end{cases}$$

(provide a proof of this)

As for the resolvent of $\mathcal{B}$, it corresponds to the projection (x,y) of a vector $(\bar{x},\bar{y})$ onto the subspace $\{(x,y) \mid [K \quad D]^T x = y\}$, expressed as

$$J_{\mathcal{B}} = (I+\lambda\mathcal{B})^{-1} = \left(x = \left(I + [K \quad D]\begin{bmatrix}K\\D\end{bmatrix}\right)^{-1}\left(\bar{x} + \begin{bmatrix}K\\D\end{bmatrix}\bar{y}\right), y = \begin{bmatrix}K\\D\end{bmatrix}x\right)$$

Algorithm 1: Primal Douglas-Rachford Splitting

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : (z_1^0, z_2^0)

for $k = 1, 2, \ldots,$ **do**
  $x^k \leftarrow \mathbf{prox}_{tf}(z_1^{k-1})$; `// Performs resolvent of` $\mathcal{A}$
  $y^k \leftarrow \mathbf{prox}_{tg}(z_2^{k-1})$;
  $u^k \leftarrow (I + A^T A)^{-1}\big(2x^k - z_1^{k-1} + A^T(2y^k - z_2^{k-1})\big)$; `// Performs resolvent of` $\mathcal{B}$
  $v^k \leftarrow A(u^k)$;
  $z_1^k \leftarrow z_1^{k-1} + \rho(u^k - x^k)$;
  $z_2^k \leftarrow z_2^{k-1} + \rho(v^k - y^k)$;
end
return $x\mathrm{Sol} = \mathbf{prox}_{tf}(z^k)$;

2.2 Primal-Dual Douglas Rachford Splitting

In this algorithm, we consider the dual problem,

$$\text{maximize}_y \; -f^*(-A^T y) - g^*(y) \tag{21}$$

with the optimality conditions for both the primal and the dual problem

$$0 \in \begin{bmatrix} 0 & 0 & A^T & I \\ 0 & 0 & -I & 0 \\ -A & I & 0 & 0 \\ -I & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g(y) \\ 0 \\ \partial f^*(w) \end{bmatrix}$$

In the above, w is the dual variable and z is the dual multiplier for the constraint $y = Ax$, thus the dual problem is subject to

$$A^T z + w = 0$$

We reduce this to a smaller system,

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

Jonathan Campana (Student ID: 260985613) We ball.

Elliot Forcier-Poirier (Student ID:) We ball.

Edmund Wu (Student ID:) We ball.

References

[1] Jim Douglas and H. H. Rachford. On the numerical solution of heat conduction problems in two and three space variables. *Transactions of the American Mathematical Society*, 82 (2):421, Jul 1956. doi: https://doi.org/10.2307/1993056. URL `https://typeset.io/pdf/`

on-the-numerical-solution-of-heat-conduction-problems-in-two-4lbdiqc5i8.pdf.

[2] Daniel O'Connor and Lieven Vandenberghe. Primal-dual decomposition by operator splitting and applications to image deblurring. *SIAM Journal on Imaging Sciences*, 7(3):1724–1754, 2014. doi: 10.1137/13094671X. URL https://doi.org/10.1137/13094671X.

[3] Neal Parikh. Proximal algorithms. *Foundations and Trends in Optimization*, 1(3):127–239, 2014. doi: https://doi.org/10.1561/2400000003.

A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.